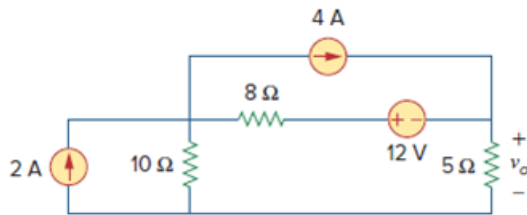


Problem

Use superposition to find v_o in the circuit of Fig. 4.81.

Figure 4.81:



Step-by-step solution

Step 1 of 13

Consider the following circuit diagram:

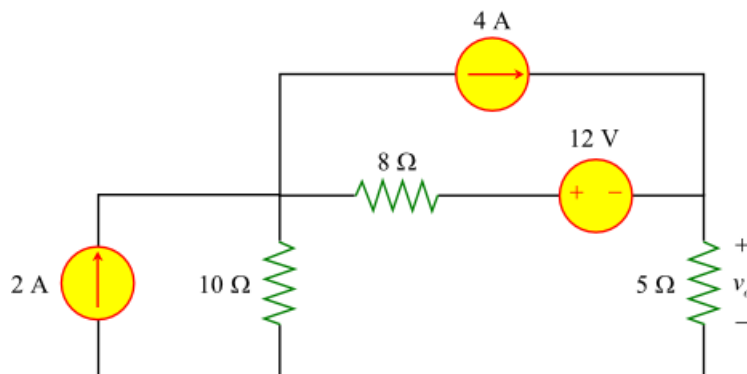


Figure 1

Step 2 of 13

The superposition theorem states that the current or voltage at any point in the linear network containing multiple independent sources may be calculated as the algebraic sum of the individual contributions of each source acting alone.

Use the principle of superposition to calculate the voltage, v_o in the circuit of Figure 1 as follows:

The following is the expression for the voltage, v_o .

$$v_o = v_{o1} + v_{o2} + v_{o3} \dots (1)$$

Here,

The voltage, v_{o1} is the contribution of the 12 V voltage source alone.

The voltage, v_{o2} is the contribution of the 2 A current source alone.

The voltage, v_{o3} is the contribution of the 4 A current source alone.

Step 3 of 13

Open circuit the 2 A and 4 A current sources and redraw the circuit in Figure 1 as follows to find the voltage, v_{o1} due to 12 V voltage source alone:

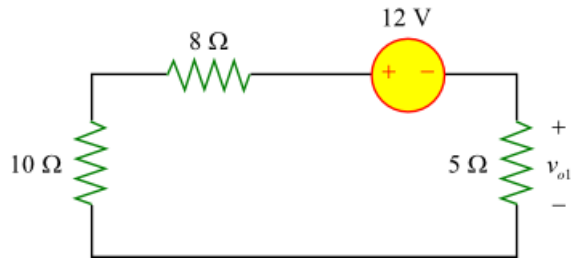


Figure 2

Step 4 of 13

Simplify the resistances in the circuit of Figure 2 as follows:

The $8\ \Omega$ and $10\ \Omega$ resistances are in series.

The following is the series combination of $8\ \Omega$ and $10\ \Omega$ resistances.

$$8\ \Omega + 10\ \Omega = 18\ \Omega$$

Step 5 of 13

Redraw the circuit in Figure 2 as follows:

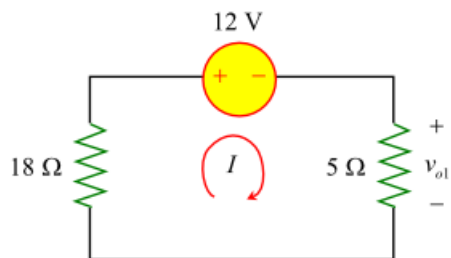


Figure 3

Step 6 of 13

The $18\ \Omega$ and $5\ \Omega$ resistances in the circuit of Figure 3 are in series.

The $12\ \text{V}$ voltage divide across the resistances, $18\ \Omega$ and $5\ \Omega$.

Use the voltage division method to find the voltage, v_{o1} .

$$\begin{aligned} -v_{o1} &= \left(\frac{5\ \Omega}{5\ \Omega + 18\ \Omega} \right) (12\ \text{V}) \\ &= \left(\frac{5\ \Omega}{23\ \Omega} \right) (12\ \text{V}) \\ &= (0.217)(12\ \text{V}) \\ &= 2.6\ \text{V} \end{aligned}$$

$$v_{o1} = -2.6\ \text{V}$$

Short circuit the $12\ \text{V}$ voltage source, open circuit the $4\ \text{A}$ current source and redraw the circuit in Figure 1 as follows to find the voltage, v_{o2} due to $2\ \text{A}$ current source alone:

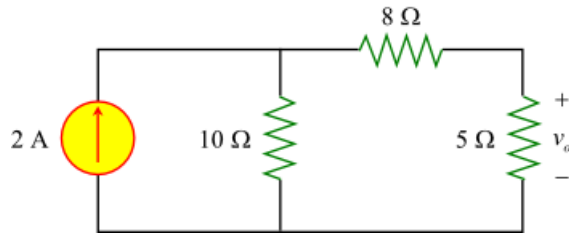


Figure 4

Step 7 of 13

Simplify the resistances in the circuit of Figure 4 as follows:

The $8\ \Omega$ and $5\ \Omega$ resistances are in series and the equivalent resistance of $8\ \Omega$ and $5\ \Omega$ resistances is in parallel with the $10\ \Omega$ resistance.

The following is the series combination of $8\ \Omega$ and $5\ \Omega$ resistances.

$$8\ \Omega + 5\ \Omega = 13\ \Omega$$

Step 8 of 13

Redraw the circuit in Figure 4 as follows:

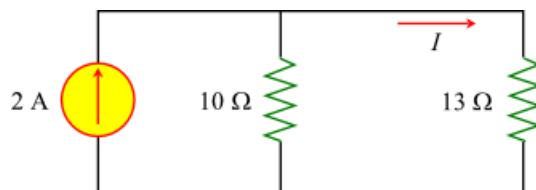


Figure 5

Step 9 of 13

Use the current division method to calculate the current, I .

$$\begin{aligned} I &= \left(\frac{10 \, \Omega}{10 \, \Omega + 13 \, \Omega} \right) (2 \, \text{A}) \\ &= \left(\frac{10 \, \Omega}{23 \, \Omega} \right) (2 \, \text{A}) \\ &= (0.435)(2 \, \text{A}) \\ &= 0.87 \, \text{A} \end{aligned}$$

The current, I flows through the series resistances, $8 \, \Omega$ and $5 \, \Omega$.

Calculate the voltage, v_{o2} as follows:

$$\begin{aligned} v_{o2} &= (5 \, \Omega) I \\ &= (5 \, \Omega)(0.87 \, \text{A}) \\ &= 4.35 \, \text{V} \end{aligned}$$

Step 10 of 13

Short circuit the 12 V voltage source, open circuit the 2 A current source and redraw the circuit in Figure 1 as follows to find the voltage, v_{o3} due to 4 A current source alone:

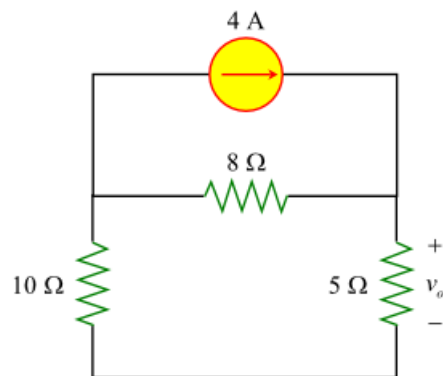


Figure 6

Step 11 of 13

Simplify the resistances in the circuit of Figure 6 as follows:

The $10 \, \Omega$ and $5 \, \Omega$ resistances are in series and the equivalent resistance of $10 \, \Omega$ and $5 \, \Omega$ resistances is in parallel with the $8 \, \Omega$ resistance.

The following is the series combination of $10 \, \Omega$ and $5 \, \Omega$ resistances.

$$10 \, \Omega + 5 \, \Omega = 15 \, \Omega$$

Step 12 of 13

Redraw the circuit in Figure 6 as follows:

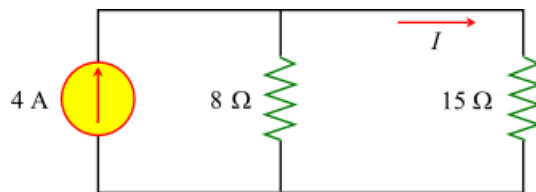


Figure 7

Use the current division method to calculate the current, I .

$$\begin{aligned} I &= \left(\frac{8 \, \Omega}{8 \, \Omega + 15 \, \Omega} \right) (4 \, \text{A}) \\ &= \left(\frac{8 \, \Omega}{23 \, \Omega} \right) (4 \, \text{A}) \\ &= (0.347) (4 \, \text{A}) \\ &= 1.39 \, \text{A} \end{aligned}$$

The current, I flows through the series resistances, $5 \, \Omega$ and $10 \, \Omega$.

Calculate the voltage, v_{o3} as follows:

$$\begin{aligned} v_{o3} &= (5 \, \Omega) I \\ &= (5 \, \Omega) (1.39 \, \text{A}) \\ &= 6.95 \, \text{V} \end{aligned}$$

Step 13 of 13

Substitute $-2.6 \, \text{V}$ for v_{o1} , $4.35 \, \text{V}$ for v_{o2} , and $6.95 \, \text{V}$ for v_{o3} in equation (1) to find the voltage, v_o .

$$\begin{aligned} v_o &= v_{o1} + v_{o2} + v_{o3} \\ &= -2.6 \, \text{V} + 4.35 \, \text{V} + 6.95 \, \text{V} \\ &= 8.7 \, \text{V} \end{aligned}$$

Thus, the voltage, v_o in the circuit of Figure 1 is 8.7 V.